15/01/2021

Aakash
Medical | IIT-JEE | Foundations

***Regd. Office*:** Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 5

Answer Key

1.	(2)	37.	(2)	73.	(1)	109.	(4)	145.	(1)
2.	(3)	38.	(1)	74.	(2)	110.	(1)	146	(2)
3.	(3)	39.	(3)	75.	(2)	111.	(2)	147.	(2)
4.	(1)	40.	(2)	76.	(1)	112.	(3)	148.	(1)
5.	(3)	41.	(1)	77.	(4)	113.	(3)	149.	(1)
6.	(1)	42.	(3)	78.	(1)	114.	(2)	150.	(2)
7.	(2)	43.	(1)	79.	(3)	115.	(1)	151.	(2)
8.	(2)	44.	(1)	80.	(4)	116.	(4)	152.	(1)
9.	(2)	45.	(2)	81.	(2)	117.	(4)	153.	(4)
10.	(4)	46.	(1)	82.	(1)	118.	(3)	154.	(2)
11.	(3)	47.	(3)	83.	(4)	119.	(4)	155.	(2)
12.	(1)	48.	(3)	84.	(3)	120.	(3)	156.	(1)
13.	(2)	49.	(2)	85.	(2)	121.	(2)	157.	(2)
14.	(4)	50.	(4)	86.	(4)	122.	(4)	158.	(2)
15.	(4)	51.	(2)	87.	(4)	123.	(1)	159.	(2)
16.	(2)	52.	(4)	88.	(3)	124.	(3)	160.	(4)
17.	(3)	53.	(2)	89.	(3)	125.	(2)	161.	(1)
18.	(1)	54.	(3)	90.	(2)	126.	(2)	162.	(3)
19.	(3)	55.	(1)	91.	(1)	127.	(3)	163.	(4)
20.	(4)	56.	(3)	92.	(2)	128.	(4)	164.	(1)
21.	(3)	57.	(4)	93.	(3)	129.	(1)	165.	(1)
22.	(4)	58.	(2)	94.	(2)	130.	(4)	166.	(1)
23.	(2)	59.	(2)	95.	(1)	131.	(2)	167.	(2)
24.	(3)	60.	(3)	96.	(3)	132.	(2)	168.	(1)
25.	(2)	61.	(4)	97.	(4)	133.	(1)	169.	(2)
26.	(3)	62.	(2)	98.	(3)	134.	(3)	170.	(3)
27.	(1)	63.	(2)	99.	(3)	135.	(4)	171.	(4)
28.	(1)	64.	(2)	100.	(2)	136.	(3)	172.	(2)
29.	(3)	65.	(4)	101.	(2)	137.	(4)	173.	(3)
30.	(2)	66.	(2)	102.	(3)	138.	(2)	174.	(2)
31.	(3)	67.	(1)	103.	(1)	139.	(1)	175.	(1)
32.	(1)	68.	(3)	104.	(4)	140.	(2)	176.	(1)
33.	(4)	69.	(1)	105.	(2)	141.	(2)	177.	(2)
34.	(2)	70.	(1)	106.	(3)	142.	(4)	178.	(3)
35.	(2)	71.	(2)	107.	(1)	143.	(4)	179.	(3)
36.	(3)	72.	(4)	108.	(1)	144.	(3)	180.	(3)

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Test - 5

Hints and Solutions

PHYSICS

1. Answer (2)

$PV = nRT$ (ideal gas)

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\therefore\ T_2 = \frac{2V}{V} \times T = 2T$

$\Delta U = \frac{f}{2} \times n \times R\Delta T$

$= 1 \times \frac{3}{2} \times R \times (2T - T) = \frac{3RT}{2}$

2. Answer (3)

$3 \times \frac{5}{2}(R)(400 - T) = 1 \times \frac{6}{2}R(T - 300)$

$15\ R\ (400 - T) = 6\ R\ (T - 300)$

$6000\ R - 15\ RT = 6\ RT - 1800\ R$

$7800\ R = 6\ RT + 15\ RT$

$\therefore\ 7800 = 21T$

$\therefore\ T = \frac{7800}{21} = \frac{2600}{7}$

$T = 371.4$ K

3. Answer (3)

$W = nRT \log_e\left(\frac{V_2}{V_1}\right) = 5 \times 8.3 \times 300 \times \log_e\left(\frac{1}{2}\right)$

$= 5 \times 8.3 \times 300 \times (-0.693)$

$= -8628$ J $= -8.6 \times 10^3$ J

4. Answer (1)

$\text{KE} = \frac{3}{2}nRT$

Temperature of the gases is same so that average translational KE of all is same.

5. Answer (3)

Final temperature equal to initial temperature $\Delta T = 0$, $\Delta U = 0$

6. Answer (1)

$\frac{dW}{dQ} = \frac{R}{C_P} = 1 - \frac{1}{\gamma} = \frac{2}{5}$

7. Answer (2)

$PV \propto T$

In compression work on system so internal energy increases and temperature rises.

$\therefore$ PV is positive

8. Answer (2)

$P \propto V^2$

$PV = nRT$

$V^2V \propto T$

$\therefore\ T \propto V^3$

If temperature increases, volume also increases and work done ($P\Delta V$) will be positive.

9. Answer (2)

$P = \frac{m}{M}\frac{RT}{V}$

$\therefore\ P = \left(\frac{mR}{MV}\right)T$

$\therefore$ Slop of $P - T$ graph $= \left(\frac{mR}{MV}\right)$

Slope $\propto m$

$\frac{\text{Slope}\,B}{\text{Slope}\,A} = \frac{3m}{m} = 3$

10. Answer (4)

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$\therefore\ \frac{T_2}{T_1} = \left(\frac{V}{8V}\right)^{5/3-1} = \left(\frac{1}{8}\right)^{2/3}$

$\frac{T_2}{300} = \left(\frac{1}{23}\right)^{2/3} = \frac{1}{4}$

$\therefore\ T_2 = \frac{300}{4} = 75\ \text{K}$

11. Answer (3)

Cycle is clockwise so work done by the gas will be positive.

Area inside the cyclic graph is work done by the gas

$W = (1 \times 10^5) \times (2 \times 10^{-3}) = +200\ \text{J}$

12. Answer (1)

$\eta = 1 - \frac{T_2}{T_1}$

$\frac{50}{100} = 1 - \frac{T_2}{T_1} = 1 - \frac{500}{T_1}$

$\therefore\ T_1 = 1000\ \text{K}$

$\eta' = 1 - \frac{T_2'}{T_1}$

$\frac{60}{100} = 1 - \frac{T_2'}{1000}$

$\therefore\ T_2' = 400\ \text{K}$

13. Answer (2)

Second law of thermodynamics by Kelvin-planck statement.

14. Answer (4)

Grahm's diffusion law $r \propto \frac{1}{\sqrt{M}}$

$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

$\frac{28.7}{7.2} = \sqrt{\frac{M_2}{M_1}}$

$\therefore\ \frac{M_2}{M_1} = \left(\frac{28.7}{7.2}\right)^2$

$\therefore\ M_2 = 2x\left(\frac{28.7}{7.2}\right)^2$

$= 31.79 = 32$

This is molecular mass of oxygen.

15. Answer (4)

$\lambda = \frac{1}{\sqrt{2}\,\pi n d^2}$

$\therefore\ \lambda \propto \frac{1}{r^2}$

$\frac{\lambda_1}{\lambda_2} = \left(\frac{3}{2}\right)^2 = \frac{9}{4} = 2.25$

16. Answer (2)

$y = \sin\omega t + \cos\omega t$

$= \sqrt{2}\left[\frac{1}{\sqrt{2}}\sin\omega t + \frac{1}{\sqrt{2}}\cos\omega t\right]$

$= \sqrt{2}\left[\sin\omega t\cos\frac{\pi}{4} + \cos\omega t\sin\frac{\pi}{4}\right]$

$y = \sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$

amplitude $= \sqrt{2}$

initial displacement $y = \sqrt{2}\sin\left(0 + \frac{\pi}{4}\right)$

$= \sqrt{2}\sin\frac{\pi}{4} = \sqrt{2} \times \frac{1}{\sqrt{2}} = 1$

$\therefore\ \frac{\text{amplitude}}{\text{displacement}} = \frac{\sqrt{2}}{1} = \sqrt{2}$

17. Answer (3)

$x = a\cos\left(\frac{2\pi}{T}t\right)$

$x = 6\cos\left(\frac{2\pi}{6} \times 8\right)$

$= 6\cos\left(\frac{8\pi}{3}\right)$

$= 6\cos\left(2\pi + \frac{2\pi}{3}\right) = 6\cos\left(\frac{2\pi}{3}\right)$

$= 6 \times \left(-\frac{1}{2}\right) = -3\,\text{cm}$

18. Answer (1)

$x = 2\sin\left(2t + \frac{\pi}{3}\right)$

Comparing it with $x = \sin(\omega t + \phi)$

$\therefore\ \phi = \frac{\pi}{3}$ rad and $\omega = 2$

$\frac{2\pi}{T} = 2 \quad \therefore\ T = \pi$ s

19. Answer (3)

At the highest point coin won't separate if

$m\omega^2 A \le mg$

$\therefore\ m\omega^2 A = mg$ (critical)

$\omega = \sqrt{\frac{g}{A}}$

$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{A}{g}} = 2\pi\sqrt{\frac{0.2}{9.8}} = \frac{2\times 22}{49} = 0.9\,\text{s}$

20. Answer (4)

The acceleration is directed towards equilibrium (mean) position.

21. Answer (3)

$v = \omega\sqrt{A^2 - x^2}$

$\therefore\ v = \sqrt{16}\sqrt{9 - x^2}$

Comparing $A^2 = 9$

$\omega = 4$

$\therefore\ A = 3$ m

$v_{max} = \omega A = 4 \times 3 = 12$ m s^{-1}

22. Answer (4)

In SHM $|F| = kx$

$\therefore\ F_{max} = KA$

$F = K\left(\frac{A}{2}\right)$

$\frac{F_{max}}{F} = 2$

$\therefore\ F = \frac{F_{max}}{2} = 5$ N

23. Answer (2)

$T = 2\pi\sqrt{\frac{m}{k}}$

$= 2\pi\sqrt{\frac{10}{3000}} = \frac{\pi}{5\sqrt{3}}$

24. Answer (3)

$T = 2\pi\sqrt{\frac{m}{k}}$

$\frac{4}{2} = \sqrt{\frac{500}{m'}}$

$T' = 2\pi\sqrt{\frac{m'}{k}}$

$\therefore\ m' = \frac{500}{4} = 125\,\text{gm}$

Removed mass = $\Delta m = m - m' = (500 - 125)$ = 375 gm

25. Answer (2)

During one complete oscillation four times potential energy is equal to kinetic energy. Therefore during half oscillation two times potential energy is equal to kinetic energy.

26. Answer (3)

$40 = \frac{1}{2} \times m \times (0.8)^2$

$\therefore\ m = 125$ kg ($\omega A = v_{max}$)

$k = m\omega^2 = 4\pi^2 mn^2$

$= 4\pi^2 \times (125) \times \left(\frac{1}{2\pi}\right)^2$

= 125 N m^{-1}

27. Answer (1)

$T = 2\pi\sqrt{\frac{L}{g}}$

$T \propto \sqrt{L}$

$\therefore\ \frac{T}{\sqrt{L}} = k$

Slope is constant graph will be a straight line

28. Answer (1)

$$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \text{ as } \frac{k}{m} >> \frac{b^2}{4m^2}$$

$$\omega' = \sqrt{\frac{k}{m}} = \sqrt{\frac{90}{0.2}}$$

$$T' = \frac{2\pi}{\omega'} = 2\pi \times \sqrt{\frac{0.2}{90}}$$

$$= 2 \times \pi \times \frac{\sqrt{2}}{30} = 0.3\,\text{s}$$

29. Answer (3)

$U = 10x\,(50\,x^2 + 10)$ erg

$U = (500\,x^2 + 100)$

$$F = -\frac{dU}{dx} = -1000\,x$$

$m\omega^2 = k = 1000$

$10 \times \omega^2 = 1000 \quad \therefore \omega = 10$ rad s^{-1}

$$n = \frac{\omega}{2\pi} = \frac{10}{2\pi} = \frac{5}{\pi}\ \text{s}^{-1}$$

30. Answer (2)

Fluid flow, they cannot take shear stress and shear modulus is zero.

31. Answer (3)

Crest to crest distance = wavelength

Time period = 0.2 s $\quad \therefore n = \frac{1}{T} = \frac{1}{0.2} = 5\text{Hz}$

$$\text{Wave velocity} = \frac{d}{t} = \frac{\lambda}{T} = \frac{2}{0.2} = 10\ \text{m s}^{-1}$$

32. Answer (1)

$$\frac{2\pi}{\lambda} = 0.1 \times 2\pi$$

$\therefore \lambda = 10$ cm

$\Delta x = 1$ cm

$$\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x = \frac{2\pi}{10} \times 1 = \frac{2 \times 180^\circ}{10} = 36^\circ$$

33. Answer (4)

At nodes zero displacement simultaneously and at antinodes maximum displacement simultaneously.

34. Answer (2)

$$n = \frac{P}{2l}\sqrt{\frac{T}{\mu}}$$

$$n' = \frac{P+1}{2l}\sqrt{\frac{T}{\mu}} \text{ dividing}$$

$$\therefore \quad \frac{n'}{n} = \frac{P+1}{P} \quad \therefore \frac{450}{400} = \frac{P+1}{P} \quad \therefore P = 8$$

$\therefore$ Fundamental = 50 Hz

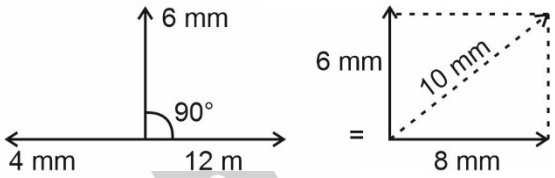
$$50 \times \frac{1}{2l} \times \sqrt{\frac{490}{0.1}} = \frac{1}{2l} \times 70$$

$\therefore \quad 100\,l = 70$

$l = 0.7$ m

35. Answer (2)

By phasor diagram



36. Answer (3)

$$v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{1.4 \times 8.31 \times 300}{2 \times 10^{-3}}}$$

$= 1320$ m s^{-1}

37. Answer (2)

(f_7) close = (f_4) open

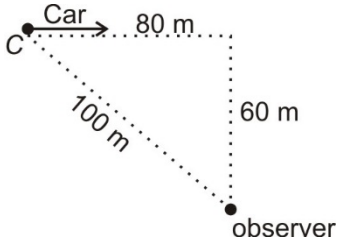
$$7 \times \frac{v}{4l_c} = 4 \times \frac{v}{2l_0}$$

$$\therefore \frac{l_c}{l_0} = \frac{7}{8}$$

38. Answer (1)

The frequency of vibration increases with increase in tension so note emitted by the string will be more than 600 Hz as it produces 5 beats per second with 600 Hz fork, the frequency will be 605 Hz.

39. Answer (3)



From geometry, $\cos\theta = \frac{80}{100} = \frac{4}{5}$

Apparent frequency = $\frac{V}{(V - V_s\cos\theta)} \times f$

$f' = \frac{330}{\left(330 - 20 \times \frac{4}{5}\right)} \times 500$ = 525.5 Hz

40. Answer (2)

Sound level $L = 10\log\frac{I}{I_0} = 60$

$\frac{I}{I_0} = 10^6$

$\therefore$ $I = I_0 \times 10^6 = 10^{-12} \times 10^6$

$= 10^{-6}$ watt m^{-2}

$= 1\ \mu W\ m^{-2}$

41. Answer (1)

For sound wave water is rarer medium and air denser.

As it travels from rarer to denser, it will bend towards normal

42. Answer (3)

$v = \sqrt{\frac{T}{\mu}}$

$\therefore$ $v \propto \sqrt{T}$ graph will be parabola.

43. Answer (1)

$3 = \frac{v}{\lambda_1} - \frac{v}{\lambda_2} = 300\left[\frac{1}{2} - \frac{1}{\lambda}\right]$

$\frac{3}{300} = \frac{1}{2} - \frac{1}{\lambda}$

$\therefore$ $\frac{1}{\lambda} = \frac{49}{100}$

$\therefore$ $\lambda = \frac{100}{49} = 2.04\,m$

44. Answer (1)

On reflection from a fixed end, there is phase change of π and velocity is reversed.

45. Answer (2)

$\mu = \frac{M}{L} = \frac{2.5}{20} = 0.125$ kg m^{-1}

$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{200}{0.125}} = 40$ m s^{-1}

$t = \frac{L}{V} = \frac{20}{40} = 0.5\,s$

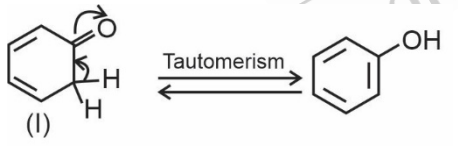
CHEMISTRY

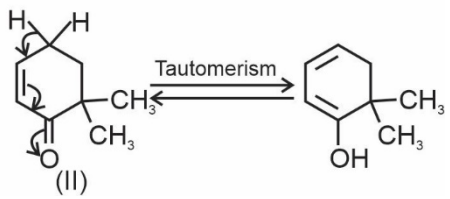
46. Answer (1)

Order of stability of carbocation is

$(CH_3)_3\overset{\oplus}{C} > Ph - \overset{\oplus}{C}H_2 > CH_2 = CH - \overset{\oplus}{C}H_2 > CH_2 = \overset{\oplus}{C}H$

47. Answer (3)





48. Answer (3)

% of nitrogen $= \frac{14 \times 10 \times 100}{1000 \times 0.2}$

= 70%

49. Answer (2)

Generally, electrophiles are Lewis acids.

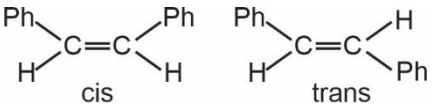
50. Answer (4)

- Greater is the stability of the alkene, lesser will be the heat of hydrogenation.
- Stability of alkene $\propto$ No. of α-hydrogen.

51. Answer (2)

2-Chloro-1-methyl-4-nitrobenzene

52. Answer (4)



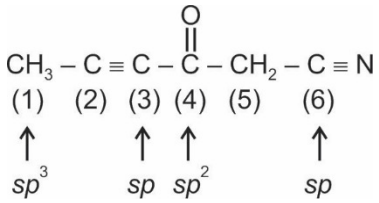
53. Answer (2)

General formula of alkanone is $C_nH_{2n}O$.

54. Answer (3)

Carboxylic acid and esters with same molecular formula are functional isomers of each other.

55. Answer (1)



56. Answer (3)

Correct order of –I effect is

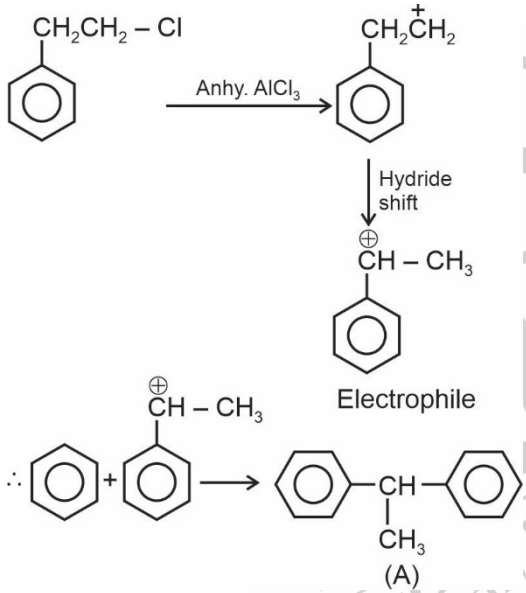
$-NO_2 > -CN > -COOH > -OH$

57. Answer (4)

 is cyclic, planar, conjugated and follows Huckel $(4n + 2)\pi$ rule.

$\therefore$ It is aromatic in nature.

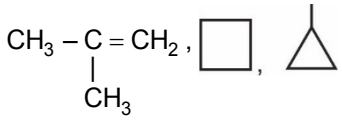
58. Answer (2)



59. Answer (2)

Degree of unsaturation is 1. Possible structural isomers are

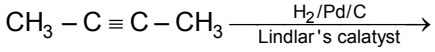
$CH_3 - CH_2 - CH = CH_2$, $CH_3 - CH = CH - CH_3$,

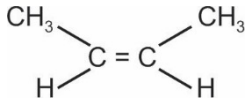


60. Answer (3)

Strong electron withdrawing group decreases the reactivity of the benzene ring towards Friedel-Crafts reaction.

61. Answer (4)

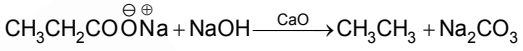




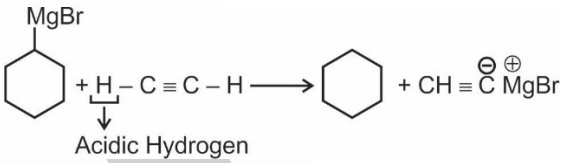
62. Answer (2)

- More is the number of covalent bonds, higher will be the stability of the resonating structure.
- Negative charge is more stable on more electronegative atom.

63. Answer (2)



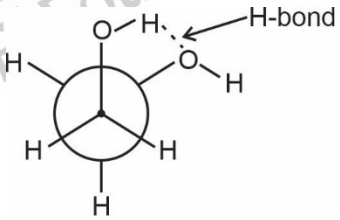
64. Answer (2)



65. Answer (4)

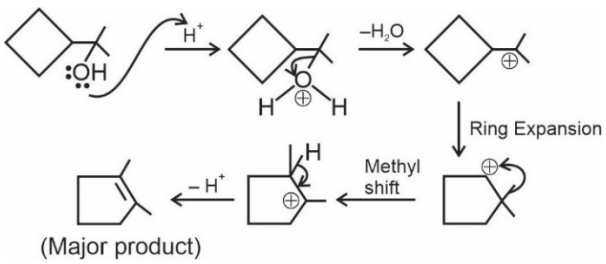
Unsaturated compounds except benzene react with bromine water.

66. Answer (2)

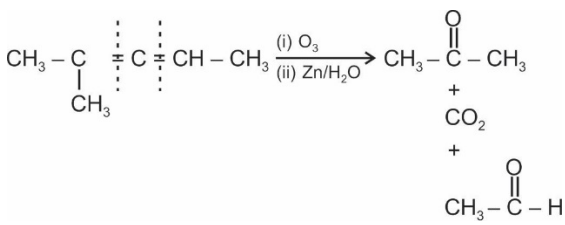


Gauche form is most stable because of intramolecular hydrogen bonding.

67. Answer (1)



68. Answer (3)



69. Answer (1)

Liquids which are having sufficient difference in boiling points are separated by distillation.

70. Answer (1)

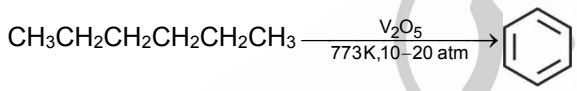
$S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}$

violet

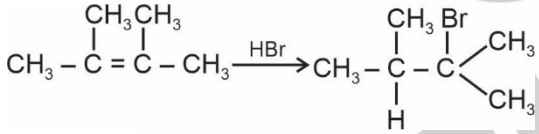
71. Answer (2)

Excess nitrates in drinking water can cause disease such as methemoglobinemia ('blue baby' syndrome).

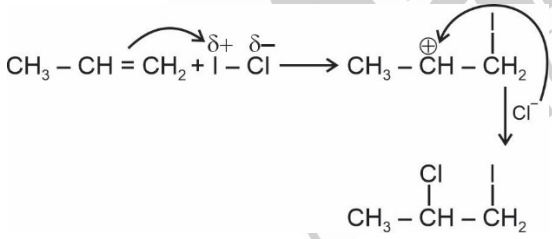
72. Answer (4)



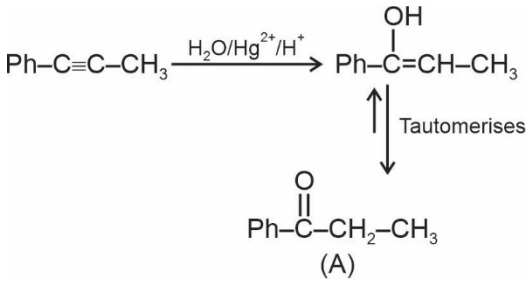
73. Answer (1)



74. Answer (2)



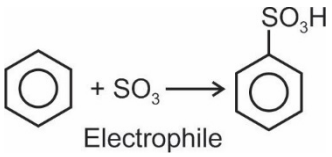
75. Answer (2)



76. Answer (1)

Lassaigne's test for nitrogen is applicable for organic compounds containing nitrogen.

77. Answer (4)



78. Answer (1)

Maximum prescribed concentration of Fe in drinking water is 0.2 ppm.

79. Answer (3)

Normally rain water has a pH 5.6.

80. Answer (4)

Methane, water vapour, nitrous oxide, CFCs, ozone and carbon-dioxide are the green house gases.

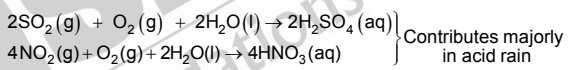
81. Answer (2)

The atmosphere that surrounds the earth is not of the same thickness at all heights.

82. Answer (1)

Higher the branching, lower is the surface area hence lesser will be van der Waals force of attraction and lower will be the boiling point.

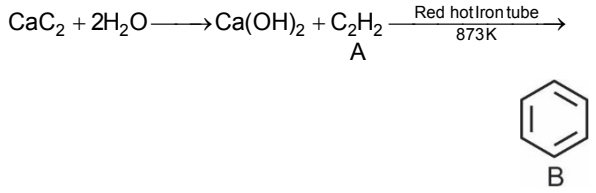
83. Answer (4)



84. Answer (3)

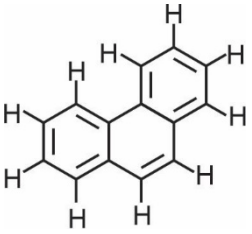
Photochemical smog has high concentration of oxidising agents and is therefore called as oxidising smog.

85. Answer (2)



86. Answer (4)

Phenanthrene is



Number of σ bonds = 26

Number of π bonds = 7

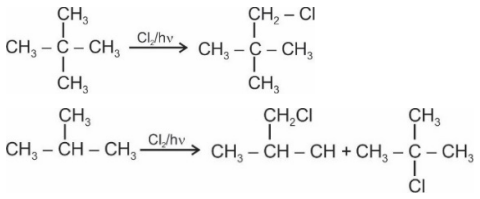
87. Answer (4)

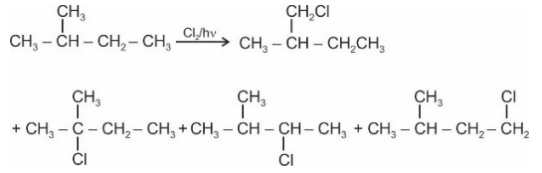
Kjeldahl's method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in the ring (e.g pyridine) as nitrogen of these compounds does not change to ammonium sulphate.

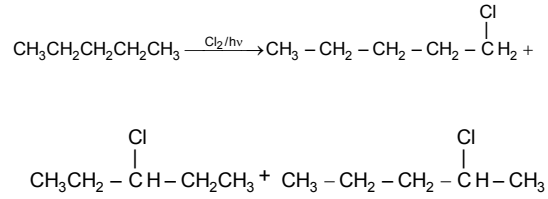
88. Answer (3)

In nucleophilic substitution reaction, one nucleophile substitutes another nucleophile.

89. Answer (3)







90. Answer (2)

Dieldrin is a pesticide.

BOTANY

91. Answer (1)

Plants unlike animals have no special systems for gaseous exchange.

92. Answer (2)

Sucrose breaks into glucose and fructose. Glucose and fructose enter glycolysis, first process of respiration.

93. Answer (3)

$2(\underset{\text{Triplamitin}}{C_{51}H_{98}O_6}) + 145O_2 \rightarrow 102CO_2 + 98H_2O$ + Energy

RQ = 102/145 = 0.7

94. Answer (2)

TCA is Krebs cycle which operates in matrix of mitochondria. (Glycolysis) and Hatch and Slack cycle operate in cytoplasm, Calvin cycle operates in stroma of chloroplast.

95. Answer (1)

In the presence of phosphotriose isomerase

Dihydroxyacetone phosphate $\rightleftharpoons$

Glyceraldehyde-3-phosphate.

96. Answer (3)

We recognise that ultimately all the food that is respired for life processes comes from photosynthesis.

97. Answer (4)

α-ketoglutaric acid is 5C molecule and succinic acid is 4C.

98. Answer (3)

In fermentation, NADH is oxidised to NAD^+ rather slowly but very vigorously in aerobic respiration.

99. Answer (3)

1 glucose molecule contains = 2870 KJ energy. It produces 38 ATP on complete oxidation.

38 ATP gives 1292 KJ energy.

Efficiency = $\frac{1292}{2870} \times 100 = 45\%$

100. Answer (2)

EMP pathway/glycolysis occurs in cytoplasm. C_3-pathway occurs in stroma of chloroplast.

101. Answer (2)

Pyruvic acid + CoA + NAD^+

$\xrightarrow[\text{Pyruvate dehydrogenase}]{Mg^{2+}}$ Acetyl CoA + CO_2 + $NADH_2$

102. Answer (3)

1 glucose molecule on glycolysis forms 2 molecules of pyruvic acid, which form 2 molecules of lactic acid on fermentation. Net gained ATP is 2.

103. Answer (1)

Three major ways are aerobic respiration, lactic acid fermentation and alcoholic fermentation.

104. Answer (4)

Less than 7 percent energy in glucose is released and not all of it is trapped as ATP.

105. Answer (2)

These element mobilise from older tissues when become deficient in soil.

106. Answer (3)

Glycolysis does not produces CO_2.

107. Answer (1)

Mineral ions are usually actively absorbed.

108. Answer (1)

Boron is absorbed in the form of BO_3^{-3} or $B_4O_7^{-2}$.

109. Answer (4)

Boron is required for pollen germination.

110. Answer (1)

Calcium is an immobile element.

111. Answer (2)

Rhodospirillum is purple non-sulphur photo-organotroph, fixes nitrogen in absence of oxygen.

112. Answer (3)

Both are free living in soil but can fix N_2 in symbiotic condition.

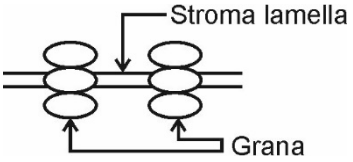
113. Answer (3)

Photosynthetic bacteria perform anoxygenic photosynthesis.

114. Answer (2)

J.V. Sachs provided evidence for the production of glucose when plants grow and it is stored in form of starch.

115. Answer (1)



116. Answer (4)

PS I = P_{700}

PS II = P_{680}

117. Answer (4)

$(2H_2O \rightarrow 4H^+ + O_2 + 4e^-) \times 4$

118. Answer (3)

First stable product of Calvin cycle is PGA, 3-carbon organic acid.

119. Answer (4)

The primary acceptor is phosphoenol pyruvic acid.

120. Answer (3)

It requires a membrane, a proton pump, a proton gradient and ATP synthase.

121. Answer (2)

Under reduction, there are a series of reactions that lead to the formation of glucose.

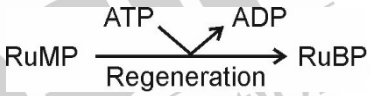
122. Answer (4)

Multilayer of large cells, without intercellular space, are found around the vascular bundles called bundle sheath cells.

123. Answer (1)

First organic acid is oxaloacetic acid. It then forms other 4-carbon compounds like malic acid or aspartic acid.

124. Answer (3)



RuMP = Ribulose monophosphate

RuBP = Ribulose bisphosphate

125. Answer (2)

Sorghum, a C_4-plant requires 5 ATP and 2NADPH to fix one CO_2 molecule.

126. Answer (2)

C_3-plants respond to higher CO_2 concentration.

127. Answer (3)

A reduced co-enzyme (NADPH) is required.

128. Answer (4)

Sodium, selenium, silicon etc are beneficial/functional elements.

129. Answer (1)

KOH soaked cotton is used for CO_2 free atmosphere.

130. Answer (4)

Complex I, complex II, complex III and complex IV.

131. Answer (2)

Hexokinase converts glucose to glucose-6-phosphate.

132. Answer (2)

Nitrogenase is made of Mo-Fe protein.

133. Answer (1)

$2NO_3 \rightarrow 2NO_2^- \rightarrow 2NO \rightarrow N_2O \rightarrow N_2\uparrow$

Denitrification process

134. Answer (3)

Different types of alcoholic drinks are obtained by distillation of the fermented broth.

135. Answer (4)

The set-up of hydroponic is costly.

ZOOLOGY

136. Answer (3)

In humans, the ciliary movement is shown by female reproductive tract to propel ova.

137. Answer (4)

The cells of cardiac muscle always assemble in a branching pattern.

138. Answer (2)

The visceral muscles assist in the transport of food through digestive tract and gametes through the genital tract by peristalsis.

139. Answer (1)

Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called fascia.

140. Answer (2)

A light zone in centre of A band is called Henson's zone which contains only thick filaments.

141. Answer (2)

The globular head along with the short arm is called heavy meromyosin (HMM) and globular head act as an ATPase enzyme.

142. Answer (4)

During muscle contraction, the size of sarcomere and H zone decreases, whereas size of overlapping band increases.

143. Answer (4)

Due to presence of plenty of mitochondria, majorly aerobic respiration occurs in red muscle fibres.

144. Answer (3)

Two zygomatic bones are present to form cheek prominences in human face.

145. Answer (1)

The number of limbs bones are 120 and the number of girdle bones are 6.

146. Answer (2)

Occipital condyles are structural constituents of occipital bones which articulate with first cervical vertebrae, atlas and forms condyloid joint.

147. Answer (2)

Each half of pectoral girdle consists of a scapula and a clavicle bone.

148. Answer (1)

Rib cage protects the visceral organs such as heart, lungs etc.

149. Answer (1)

Each pelvic girdle consists of 2 coxal bones and each coxal bone is formed by fusion of 3 bones namely ilium, ischium and pubis.

150. Answer (2)

Pivot joint is present between atlas and axis.

151. Answer (2)

All synovial joints allow considerable movements and play a significant role in locomotion.

152. Answer (1)

Muscular dystrophy is a progressive degeneration of skeletal muscles.

153. Answer (4)

Femur, tibia, fibula and tarsals are present in lower limbs.

154. Answer (2)

False ribs (floating ribs) are also attached dorsally to thoracic vertebrae.

155. Answer (2)

Each rib has two articulation surfaces on its dorsal end and is hence called bicephalic.

156. Answer (1)

Acromion process is a structural constituent of scapula.

157. Answer (2)

Vertebral column extends from the base of the skull and constitutes the main framework of the trunk.

158. Answer (2)

Cranium contains single ethmoid bone and single sphenoid bone.

159. Answer (2)

Each coxal bone of pelvic girdle is formed by fusion of three bones namely ilium, ischium and pubis.

160. Answer (4)

Clavicle articulates with scapula and also with sternum.

161. Answer (1)

Resting membrane potential is comparatively more permeable to K^+ and almost impermeable to Na^+.

162. Answer (3)

Cerebral aqueduct connects diocoel to metacoel.

163. Answer (4)

Cerebellum is not included in limbic system.

164. Answer (1)

Auditory area is present in cerebral cortex.

165. Answer (1)

Interneurons are not involved in knee jerk reflex between afferent and efferent neuron.

166. Answer (1)

In each cerebral hemisphere, the white matter is surrounded by the grey matter.

167. Answer (2)

The ear ossicle malleus is found attach with tympanum / ear drum.

168. Answer (1)

Reissner's membrane forms boundary between scala vestibule and scala media.

169. Answer (2)

Visual cortex is present in cerebral cortex whereas arachnoid is a middle menix.

170. Answer (3)

Midbrain, pons and medulla oblongata constitutes the brain stem.

171. Answer (4)

Thalamus is a major coordinating centre for sensory and motor signaling.

172. Answer (2)

Schwann cells are exclusively present in PNS.

173. Answer (3)

Impulses travel from CNS to viscera and from viscera to CNS in visceral neural system.

174. Answer (2)

It does not influence the adrenal cortex because adrenal cortex is an endocrine gland.

175. Answer (1)

Visual activity is zero at blind spot because it does not contain any photoreceptor cells.

176. Answer (1)

Crista ampullaris arises from ampulla of semi-circular canals.

177. Answer (2)

At blind spot the optic nerves leave the eye.

178. Answer (3)

Vitreous chamber is filled with vitreous humor which is a transparent gel.

179. Answer (3)

The diameter of pupil is regulated by the muscle fibres of Iris.

180. Answer (3)

Dendrites are not present in unipolar neuron.

❑ ❑ ❑

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Edition: 2020-21

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